

Sana'a UNIVERSITY



FACULTY OF ENGINEERING

Mechatronics Department

Second Year

Report Theorems

Supervision

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Mechatronics (Parallel)

Group (3)

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➤ **Introduction:**

The experience to learn what is the theorems and how we can make it and from this experience we make a one voltage and three resistances to (complete the circuit. And also we use the potentiometer resistance to know the current

➤ **Objectives:**

Measure how much the current and voltage move into some resistances at the theorems and read the measure in the (Multi meter) and how to make it

➤ **Methodology:**

From this experience we can measure the current and voltage for some resistance in Same circuit with 9v in thermal circuit

➤ **Equipment:**

Multimeter (DMM)

Tow batter (9V)

Some resistances.

Some wires.

Mother board

Potentiometer resentence

➤ Procedures:

First: put the resistance in the mother board and measure without current

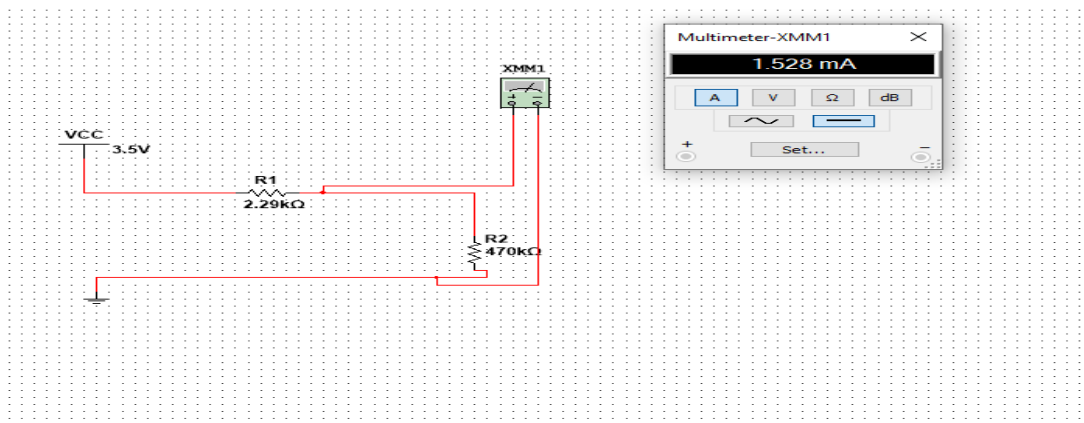
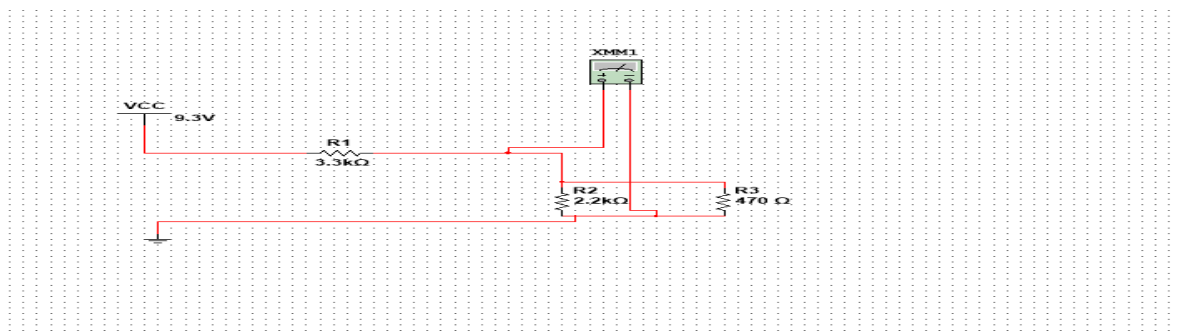
Second: connect one volt in the mother board after that connect the another volt.

Third: use one mlti meter to measure the current withe resistance with (volt1, volt2, and the both).

➤ Simulation

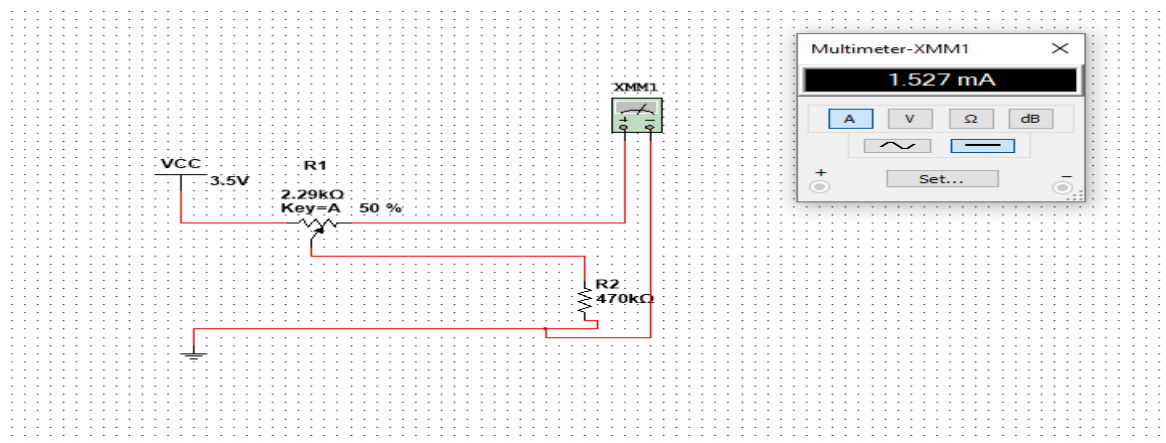
Part(1)

This simulation for measure the current



Part (2)

Potential



➤ Results:

Part (1)

$$R_1 = 3.26 \text{ k}\Omega$$

$$R_2 = 2.17 \text{ k}\Omega$$

$$R_3 = 0.99 \text{ k}\Omega$$

By calculate: $R_{th} = 2.32 \text{ k}\Omega$

$$E_{th} = 3.5 \text{ V}$$

$$I_l = 1.26 \text{ mA}$$

$$R = 1470 * 2200 / (1470 + 2200) = 881.19 \Omega$$

$$881.19 + 3300 = 41.18 \text{ k}\Omega = R_t = 4.18 \text{ k}\Omega$$

$$I_s = 9.3 / 4.8 = 2.22 \text{ mA}$$

$$I = I_s R_2 / (R_2 + R_3 // R_4) = 2.21 \text{ mA}$$

Part(2)

Poteneial

$$I_l = 2.22 (2.17 / (1.47 + 2.17)) = 1.32 \text{ mA}$$

➤ **Discussion:**

- 1) We find some problem how to do it
- 2) The second thing when we use the mutism
- 3) When we change the potential r

➤ **References:**

Electronic Circuits by Boylestad.